GGG-H-496c
FEBRUARY 6, 1961
SUPERSEDING
Int. Fed. Spec. GGG–H–00496b(GSA–FSS)
August 4, 1959 and
Fed. Spec. GGG–H–496
April 29, 1941

FEDERAL SPECIFICATION

HOE; GARDEN, MORTAR, AND MORTAR MIXING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—The hoes covered by this specification are provided with handles. The type I hoe is used for gardening and the types II and III hoes are used for mixing mortar. The type III hoe is provided with two holes in the blade to aid in the mortar-mixing operation.

1.1.1 *Federal specification coverage.*—Federal specifications do not cover all types of the commodity indicated by the titles of the specifications, but include only those most generally used by the Federal Government.

1.2 **Classification.**

1.2.1 *Types.*—The hoes covered by this specification shall be of the following types, as specified (see 6.2):

Type I. —Garden.

Type II. —Mortar.

Type III.—Mortar mixing.

1.2.2 *Sizes.*—The hoes shall be of the sizes shown in tables I, II, and III as specified by size number in the invitation to bid, contract, or order (see 6.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 **Specifications and standards.**—The following specifications and standards, of the issues in effect on date of invitation for bid, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards,

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and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of hoes otherwise in accordance with this specification.

3.2 Material.

3.2.1 *Blades, shanks, and sockets.*—Blades, shanks, and sockets shall be free from rust or defects affecting appearance or which may affect serviceability. Blades and shanks shall be made from forged steel (see 3.8). The portion of the tapered socket into which the handle is inserted shall be made of steel suitable for welding to the gooseneck shank.

3.2.2 *Handles.*—Handles shall be free from knurls, decay, holes, or other defects affecting appearance, or which may affect serviceability.

3.3 Design and construction.—Hoes shall consist essentially of a blade, gooseneck shank, crosspin, handle, and either a tapered socket or steel ferrule at the option of the contractor. The blade and the shank shall be either forged from one piece of steel, or two pieces of steel welded together so as to provide a hoe of a strength at least equal to the one-piece construction.

3.3.1 *Tapered-socket construction.* — The tapered socket shall be properly welded to the gooseneck shank, and the wall thickness of the tapered socket shall be not less than 0.060 inch.

3.3.2 *Inserted shank construction.*— The gooseneck shank shall terminate in a drive tang which shall be forced into the end of the handle at least 3 inches. Type I hoes shall be reinforced by a tapered tubular ferrule not less than 3¾ inches long. Type II and type III hoes shall be reinforced by a tapered tubular steel ferrule not less than 6 inches long; and the small end shall be protected by a steel cap or the ferrule shall be cupped. Longitudinal seams of all ferrules shall be closed by a continuous weld.

3.4 Finish and pitch of blades. — Blades shall be flat or slightly concaved, and shall have a pitch of 70 to 85 degrees with the handle. Type I hoes shall have at least a ⅓ polish, that is ⅓ of the blade near the hoeing edge shall be polished and the remainder shall be smooth finished by sandblasting, tumbling or similar operation. Types II and III hoes shall have a smooth finish produced by sandblasting, tumbling or similar operations.

3.5 Handles.—Handles shall be made either of sound thoroughly seasoned straight-grained ash or hickory. Ash handles shall be at least fair weight (not less than 30 pounds per cubic foot), and shall be all white (sapwood), all red (heartwood), mixed white and red (sapwood and heartwood), or variegated color (calico ash). Handles shall have one end rounded and the other end properly tapered to securely fit in the hoe socket or ferrule. The largest diameter on the handle shall be not less than 1 3/16 inches for type I hoes and not less than 1 5/16 inches for type II and type III hoes. Handles shall be properly shaped, smoothly finished, and clear-lacquered. There shall be not less than 3 inches

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of the handle within the socket. The length of handle shall include that part within the socket.

3.6 **Crosspins.**—Nails and rivets shall have a body diameter not less than 0.110 inch.

3.6.1 *Tapered-socket construction.*—If the hoe is of the tapered-socket construction, the handle shall be secured in the socket by means of one or more rivets or one or more well-driven nails. Nail(s), if used, shall pass through one wall of the socket and at least ⅔ the distance through the handle.

3.6.2 *Inserted-shank construction.*—If the hoe is of the inserted-shank construction the ferrule shall be secured to the handle with one or more rivets or one or more well-driven nails. Nail(s), if used, shall have at least one nail passing through one wall of the ferrule and into the inserted shank. Rivet(s), if used, shall have at least one rivet passing through the ferrule, handle, and inserted shank.

3.7 **Marking.** — The handle and/or blade shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.8 **Hardness of blade.**—Blades shall show a hardness of not less than Rockwell C-26 when tested ½ inch from the cutting edge of the blade (see 4.5.1).

3.9 **Strength tests.** — Breaking, cracking, or permanent set (in excess of ½ inch for type I hoes and ⅛ inch for type II and type III hoes) of the handle, blade or shank, when tested as outlined in 4.5.2 and 4.5.2.1, will be considered as failing to meet specification requirements.

3.10 **Socket and handle test.** — Rupture, pulling the handle out of the socket, or set in excess of ⅛ inch when tested as outlined in 4.5.3 will be considered as failing to meet specification requirements.

3.11 **Type I, garden hoes.**—Type I hoes shall conform to the dimensional requirements in table I. Garden hoes shall have a beveled edge suitable for sharpening (see figure 1).

TABLE I.—*Type I, garden hoes*

Size No.	Width of blade		Depth of blade		Length of handle		Diameter of shank
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*
1	5¾	6¼	4¼	4¾	51	54	7/16
2	6¾	7¼	4¼	4¾	51	54	7/16
3	7¾	8¼	4⅛	5	51	54	7/16

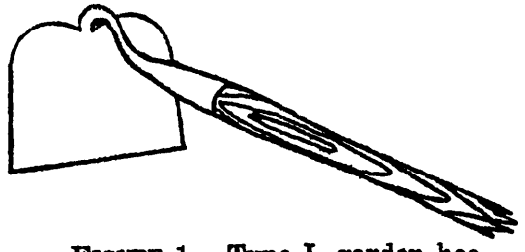


FIGURE 1.—Type I, garden hoe.

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3.12 Type II, mortar hoes.—Type II hoes shall conform to the dimensional requirements of table II. Mortar hoes shall have a sheared or square edge and be similar to figure 2.

3.13 Type III, mortar-mixing hoes.—Type III hoes shall have two round holes cut in the blades, not less than $1\frac{1}{16}$ inches in diameter; each hole shall be located approximately in the center of each half of the blade. Hoes shall have a sheared or square edge and be similar to figure 3.

3.14 Workmanship. — Workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 *Sampling for visual and dimensional inspection.*—A random sample of hoes shall be taken by the inspector in accordance with table IV from each inspection lot of material offered for inspection of visual and dimensional characteristics, with lot acceptance based on sampling requirements in accordance with Military Standard MIL–STD–105.

4.2.3 *Sampling for lot acceptance tests.*—A random sample of hoes shall be taken from each inspection lot in accordance with table V and shall be subjected to the tests specified in 4.5.

TABLE II.—*Type II, mortar hoes*

Size No.	Width of blade		Depth of blade		Length of handle		Diameter of shank
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*
1	8¾	9¼	5¾	6¼	65	67	⅝
2	9¾	10¼	5¾	6¼	65	67	⅝

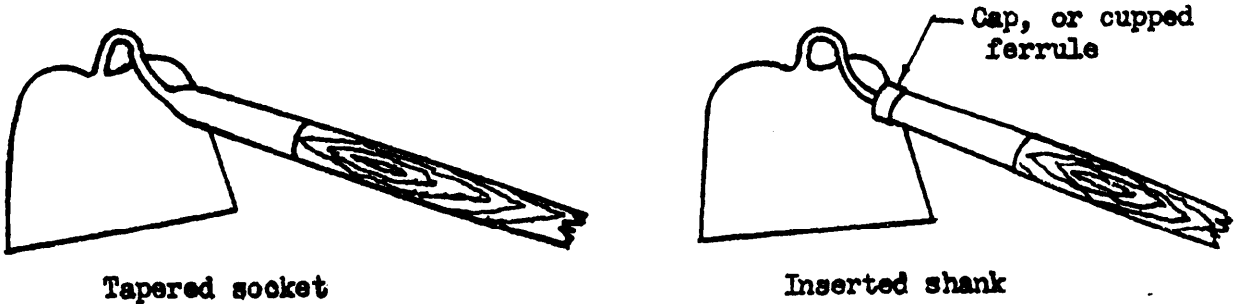


FIGURE 2.—Type II, motar hoe.

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TABLE III.—*Type III, mortar mixing hoes*

Size No.	Width of blade		Depth of blade		Length of handle		Diameter of shank
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*
1	8¾	9¼	5¾	6¼	65	67	⅝
2	9¾	10¼	5¾	6¼	65	67	⅝

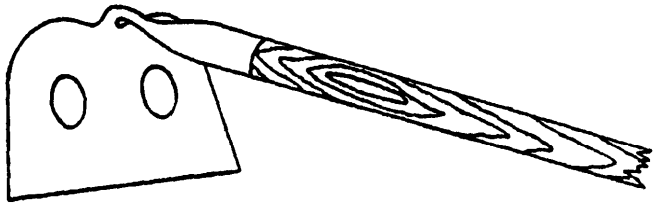


FIGURE 3.—Type III, mortar-mixing hoe. (Tapered socket shown.) (Tapered socket or inserted shank construction permitted.)

TABLE IV.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

Number of hoes in inspection lot	Number of hoes in sample	Acceptance number (defective)	Rejection number (defective)
65 and under	10	0	1
66 to 110	35	1	2
111 to 180	50	2	3
181 to 300	75	3	4
301 to 500	110	4	5
501 to 1300	150	5	6
1301 and over	225	8	9

Note.—The Government inspector may institute tightened or reduced inspection in accordance with section 9 of Military Standard MIL–STD–105.

4.3 Visual and dimensional inspection.—Each of the sample hoes taken in accordance with table IV shall be visually and dimensionally inspected by the inspector to verify compliance with this specification. The width of the blade shall be measured at the cutting edge. The depth of the blade shall be the distance between the cutting edge and the highest portion of the blade proper (excluding the shank). Any hoe in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective hoes in any sample exceeds the acceptance number for that sample, the lo represented by the sample shall be rejected.

TABLE V.—*Sampling for lot acceptance tests*

Number of hoes in inspection lot	Number of hoes in sample	Acceptance number defectives	Rejection number defectives
15 and under	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 800	15	1	2
801 to 1300	25	1	2
1301 and over	35	2	3

4.4 Lot acceptance tests.—Each sample hoe taken in accordance with table V shall be tested in accordance with the tests in 4.5 through 4.5.3. Any sample hoe failing to meet any of the tests shall be rejected and if the

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number of defective hoes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 *Hardness of blade.*—Hardness shall be determined at two or more points with a Rockwell tester with a diamond penetrator and a major load of 150 kg. (see 3.8). The parts to be tested should be firmly supported during the test.

4.5.2 *Strength tests.*—The handle shall be rigidly held in a vertical position with the blade end down, between two parallel jaws approximately 10 inches from the blade, and a load of not less than 75 pounds nor more than 80 pounds is applied over a length of 1¼ inches along the edge at approximately equal distances from the sides of the blade for a period of 2 minutes. The edge of the hoe shall be in horizontal position when the test is started.

4.5.2.1 The blade shall be held between two (parallel) jaws approximately 2 inches wide by 1 inch depth at a distance of 2 inches from the cutting edge of the blade, and at equal distances from the sides of the blade, and a load of not less than 10 pounds nor more than 12 pounds applied on the handle at a distance of 2 feet from the cutting edge of the blade, for a period of 5 minutes. A vise may be used for this test. The clamping faces of the jaws shall be vertical and the top edges shall be horizontal (see 3.9).

4.5.3 *Socket and handle test.*—The handle of the hoe shall be securely clamped in a vertical position with the blade end down, and a load of not less than 125 pounds nor more than 130 pounds applied for 5 minutes to the shank in a vertical direction. To determine set, the distance from the top end of the handle to the top edge of the socket shall be measured before and after the test. The difference between these distances is the set (see 3.10).

4.6 Rejected lots.—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming hoes. The inspector shall again examine samples from such resubmitted lots to verify compliance with the requirements of this specification.

5. PREPARATION FOR DELIVERY

(Civil agencies. — Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.

5.1.1 *Level A.*—Level A preservation and packaging shall be in accordance with level A in Military Specification MIL–P–15424.

5.1.2 *Level C.*—Level C preservation and packaging may be the supplier's commercial practice. Unless otherwise specified, hoes shall be tied in bundles of ½ dozen or 1 dozen each.

5.2 Packing.

5.2.1 *Level A.*—Level A packing shall be in accordance with level A in Military Specification MIL–P–15424.

5.2.2 *Level B.*—Level B packing shall be in accordance with level B in Military Specification MIL–P–15424.

5.2.3 *Level C.*—Level C packing shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking specified in the contract or order, shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* — In addition to any special marking specified in the contract

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or order, shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. — Type I hoes are intended for garden use. Type II and type III hoes are intended for use in mixing mortar.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Type, and size number required (see 1.2.1, 1.2.2 and tables I, II and III).

(*c*) State levels of preservation, packaging and packing required (see 5.1 and 5.2).

(*d*) Marking, if different (see 5.3).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Transportation description. — Transportation description and minimum weights applicable to this commodity are:

Rail:

Hoes, not wheeled.

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Hoes, not wheeled.

Motor volume minimum weight 24,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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